

Logical Aspects of Multi-Agent Systems

Homework set D

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BOK2 2.1

- a** We have to show that $(M, w_1) \models C(p \vee q)$. By definition of C , this means we have to show that $(M, t) \models p \vee q$ for all worlds t reachable from w_1 in any number of steps with any combination of agents (i.e., t with $s \rightarrow t$).

We can first check which worlds are reachable. World w_1 is reachable from itself, world w_2 is reachable from w_1 via R_1 and R_2 . World w_3 is reachable from w_1 via R_2 . Thus, the set of reachable worlds is $\{w_1, w_2, w_3\}$.

We need to check the truth of $p \vee q$ in all reachable worlds.

At w_1 , we have that $(M, w_1) \models p$ and $(M, w_1) \models q$, so $(M, w_1) \models p \vee q$.

At w_2 , we have that $(M, w_2) \models p$, so $(M, w_2) \models p \vee q$.

At w_3 , we have that $(M, w_3) \models q$, so $(M, w_3) \models p \vee q$.

Since $p \vee q$ is true in all worlds reachable from w_1 , we conclude $(M, w_1) \models C(p \vee q)$.

- b** We have to show that $(M, w_1) \models C(p \vee q) \rightarrow (Cp \vee Cq)$. This is an implication. It is false if and only if the antecedent $(C(p \vee q))$ is true and the succedent $((Cp \vee Cq))$ is false.

From part **a**, we already established that $(M, w_1) \models C(p \vee q)$ is true.

Now we must evaluate the succedent: $(Cp \vee Cq)$. This requires checking if $(M, w_1) \models Cp$ or $(M, w_1) \models Cq$.

First, we check $(M, w_1) \models Cp$. The set of reachable worlds from w_1 is $\{w_1, w_2, w_3\}$ as explained above. At w_3 , we have $(M, w_3) \models \neg p$. Since p is not true in all reachable worlds, $(M, w_1) \not\models Cp$.

Next, check Cq . Again, we check the reachable worlds. At w_2 , we have $(M, w_2) \models \neg q$. Since q is not true in all reachable worlds, $(M, w_1) \not\models Cq$.

Since both Cp and Cq are false, the disjunction $(Cp \vee Cq)$ is false.

Because the antecedent is true and the succedent is false, we conclude that $(M, w_1) \not\models C(p \vee q) \rightarrow (Cp \vee Cq)$.

- c** We have to determine the truth value of $(M, w_1) \models C(p \wedge q)$. By definition of C , this means we have to show that $(M, t) \models p \wedge q$ for all worlds t reachable from w_1 .

As established, the set of reachable worlds is $\{w_1, w_2, w_3\}$. We need to check the truth of $p \wedge q$ in these worlds.

At w_1 , we have $(M, w_1) \models p$ and $(M, w_1) \models q$, so $(M, w_1) \models p \wedge q$.

However, at w_2 , we have $(M, w_2) \models \neg q$. Thus, $(M, w_2) \not\models p \wedge q$.

Since $p \wedge q$ is not true in all reachable worlds, we conclude that $(M, w_1) \not\models C(p \wedge q)$.

- d** We need to show that $(M, w_1) \models C(p \wedge q) \leftrightarrow (Cp \wedge Cq)$. This is the same as showing that $(M, w_1) \models (C(p \wedge q) \rightarrow (Cp \wedge Cq)) \wedge ((Cp \wedge Cq) \rightarrow C(p \wedge q))$.

We can first show $(M, w_1) \models C(p \wedge q) \rightarrow (Cp \wedge Cq)$.

This is the same as showing $(M, w_1) \not\models C(p \wedge q)$ or $(M, w_1) \models Cp \wedge Cq$.

We know from **c** that $(M, w_1) \not\models C(p \wedge q)$.

Thus, we know that $(M, w_1) \not\models C(p \wedge q)$ or $(M, w_1) \models Cp \wedge Cq$.

Thus, we know that $(M, w_1) \models C(p \wedge q) \rightarrow (Cp \wedge Cq)$.

We can now show the second part of the conjunction. That is, $(M, w_1) \models (Cp \wedge Cq) \rightarrow C(p \wedge q)$.

This can be rewritten as $(M, w_1) \not\models Cp \wedge Cq$ or $(M, w_1) \models C(p \wedge q)$.

We evaluate $Cp \wedge Cq$. First, check Cp . The set of reachable worlds is $\{w_1, w_2, w_3\}$. At w_3 , we have $(M, w_3) \models \neg p$. Thus $(M, w_1) \not\models Cp$.

Since Cp is false, the conjunction $(Cp \wedge Cq)$ is false.

Thus, we have $(M, w_1) \not\models Cp \wedge Cq$. Thus, we have that $(M, w_1) \models (Cp \wedge Cq) \rightarrow C(p \wedge q)$.

We have shown that both parts of the conjunction are true, i.e., $(M, w_1) \models (C(p \wedge q) \rightarrow (Cp \wedge Cq)) \wedge ((Cp \wedge Cq) \rightarrow C(p \wedge q))$. Thus, we conclude $(M, w_1) \models C(p \wedge q) \leftrightarrow (Cp \wedge Cq)$.

- e** We have to determine the truth value of $(M, w_1) \models C(p \vee q) \rightarrow CC(p \vee q)$.

This is the same as saying $(M, w_1) \not\models C(p \vee q)$ or $(M, w_1) \models CC(p \vee q)$.

First, evaluate the antecedent $C(p \vee q)$. From part **a**, we already established that this is true: $(M, w_1) \models C(p \vee q)$.

Now, evaluate the succedent $CC(p \vee q)$. By definition of C , this means determining if $C(p \vee q)$ holds in all worlds reachable from w_1 (i.e., $\{w_1, w_2, w_3\}$).

We check the status of $C(p \vee q)$ at each world:

- At w_1 : from **a** we know $(M, w_1) \models C(p \vee q)$.
- At w_2 : The reachable worlds are $\{w_1, w_2, w_3\}$. In all of them, $p \vee q$ holds (through similar reasoning as shown in **a**). So $(M, w_2) \models C(p \vee q)$.

- At w_3 : The reachable worlds are $\{w_1, w_2, w_3\}$. In all of them, $p \vee q$ holds (through similar reasoning as shown in **a**). So $(M, w_3) \models C(p \vee q)$.

Since $C(p \vee q)$ is true in all reachable worlds, $(M, w_1) \models CC(p \vee q)$ is true. Thus, we have that $(M, w_1) \models C(p \vee w) \rightarrow CC(p \vee q)$.

- f** We have to determine the truth value of $(M, w_1) \models \neg Cp \rightarrow C\neg Cp$.

This is the same as checking $(M, w_1) \models \neg\neg Cp \vee C\neg Cp$.

This is the same as checking $(M, w_1) \models Cp$ or $(M, w_1) \models C\neg Cp$.

We check the first part of the disjunction, $(M, w_1) \models Cp$. The set of reachable worlds is $\{w_1, w_2, w_3\}$. At w_3 , we have $(M, w_3) \models \neg p$. Thus $(M, w_1) \not\models Cp$.

Now, evaluate the second part of the disjunction $C\neg Cp$. By definition, this means we must show that $\neg Cp$ holds in all worlds reachable from w_1 (i.e., we must show that Cp fails in every reachable world).

- At w_1 : Reachable worlds are $\{w_1, w_2, w_3\}$. p fails at w_3 . Thus $(M, w_1) \models \neg Cp$.
- At w_2 : Reachable worlds are $\{w_1, w_2, w_3\}$. p fails at w_3 . Thus $(M, w_2) \models \neg Cp$.
- At w_3 : Reachable worlds are $\{w_1, w_2, w_3\}$. p fails at w_3 . Thus $(M, w_3) \models \neg Cp$.

Since $\neg Cp$ is true in all reachable worlds, we have that $(M, w_1) \models C\neg Cp$.

Thus, we have that $(M, w_1) \models \neg\neg Cp \vee C\neg Cp$. Thus, we have that $(M, w_1) \models \neg Cp \rightarrow C\neg Cp$.

BOK2 2.2

- a** Let $\varphi = p$. We can have a truth assignment π so that p and q are true in all worlds $w_1, w_2, \dots, w_7$, simply by setting $\pi(w_i)(p) = 1$ for $i \in \{1, \dots, 7\}$ and $\pi(w_i)(q) = 1$ for $i \in \{1, \dots, 7\}$.

At any world of the seven worlds, we can see that Cp and E^7p are true.

Thus, we have that both $M \models E^7p \rightarrow Cp$ and $M \models Cp \rightarrow E^7p$.

Thus, we have that $M \models E^7p \leftrightarrow Cp$ in this model.

- b** To prove that $M \models E^n\varphi \leftrightarrow Cp$ holds in all $S5_n$ Kripke models M with n states, we need to show that for any state s in a model with n states, we have $(M, s) \models E^n\varphi \leftrightarrow Cp$.

First, we show the trivial ($\leftarrow$) direction: $(M, s) \models C\varphi \rightarrow E^n\varphi$.

By definition, common knowledge implies that everyone knows φ at any depth $E^k\varphi$ for all $k \geq 1$.

Thus, if $C\varphi$ is true, $E^n\varphi$ is automatically true.

Thus, we have $(M, s) \models C\varphi \rightarrow E^n\varphi$.

We now show the $(\rightarrow)$ direction: $(M, s) \models E^n\varphi \rightarrow C\varphi$.

By the truth definition of C , $C\varphi$ holds at a state s if φ is true in all worlds t reachable from s .

In a graph with n states, the longest path between any two connected nodes is at most $n - 1$.

Since the maximum distance is $n - 1$, the operator E^n covers every reachable node from any state s . Thus, if $E^n\varphi$ is true (meaning φ is true in all worlds within distance n), then φ is true in all reachable worlds. This is the definition of $C\varphi$.

Thus, we have $(M, s) \models E^n\varphi \rightarrow C\varphi$.

Thus, we have that $(M, s) \models E^n\varphi \leftrightarrow C\varphi$.

Because s was picked arbitrarily, we have that $M \models E^n\varphi \leftrightarrow C\varphi$.

- c Let $\varphi = p$. For the truth assignment π' we should have that p is false at world w_1 and true everywhere else, so $\pi'(w_1)(p) = 0$ and $\pi'(w_i)(p) = 1$ for all i strictly greater than 1.

Since the graph is connected, w_1 is reachable from any world w_k through a combination of agents. Because p is false at w_1 , it is not the case that p is true at all reachable worlds k . Therefore, Cp is false at every world.

We need to find a world where $E^n p$ is true. We can choose world w_{n+2} . From w_{n+2} , the reach of the E^n operator includes all worlds within a distance of n . Thus, the lowest index reachable from w_{n+2} within n steps is 2. The set of worlds checked by E^n at w_{n+2} is thus $\{w_2, w_3, \dots, w_{2n+2}\}$. According to the truth assignment, p is true in all of these worlds. Therefore, we have at world w_{n+2} for any $n \geq 0$ that $(M, w_{n+2}) \models E^n p$ is true while $(M, w_{n+2}) \not\models Cp$. Thus, we have that $(M, w_{n+2}) \not\models E^n p \leftrightarrow Cp$. As we have this for any world w_n with $n \geq 0$, this means that $M \not\models E^n p \leftrightarrow Cp$.

BOK2 2.4

We need to show that for every formula φ with an arbitrary length $n \in \mathbb{N}$ (though finite), the closure Φ of φ with respect to $\mathbf{KEC}_{(m)}$ is a finite set of formulas.

We will go through all of the 5 steps that are used to make the closure of φ as stated in the definition.

For $a.$, we have $\varphi \in \Phi$. As φ has a finite length, adding this will simply increase one formula.

For $b.$, we have that if $\Psi \in \Phi$ and χ is a subformula of Ψ , then $\chi \in \Phi$.

Since any formula Ψ has finite length, adding the subformulas χ to Φ means adding a finite number of formulas to Φ . Any further subformulas of χ will

similarly be of finite length, and thus a finite number of formulas will be added to Φ in this step.

For $c.$, we have that if $\Psi \in \Phi$ and Ψ itself is not a negation, then $\neg\Psi \in \Phi$.

This will simply add one formula to Φ , being $\neg\Psi$. Once we have $\neg\Psi \in \Phi$, we will not use this step anymore because it will be a negation.

For $d.$, we have that if $C\Psi \in \Phi$ then $EC\Psi \in \Phi$.

This will similarly add just one formula to Φ , being $EC\Psi$. Any subformula from Ψ was already in the set from step b . There is thus no loop.

For $e.$, we have that if $E\Psi \in \Phi$ then $K_i\Psi \in \Phi$ for all $i \leq m$. Since m is a finite number, there will be a finite number of formulas added to Φ . We also don't have to worry about infinite loops, since any Ψ will already be a subformula and thus have been put into the set at step b .

In conclusion, we can see that at any step, a finite number of formulas is added to Φ . Furthermore, we see that there are no infinite loops (just finite ones) between steps. Thus, we can see that for every formula φ with an arbitrary length $n \in \mathbb{N}$ (though finite), the closure Φ of φ with respect to $\mathbf{KEC}_{(m)}$ is a finite set of formulas.

BOK2 2.5

a Γ_1 is not a maximally $\mathbf{KEC}_{(m)}$ -consistent set in Φ . This is because Cp and $\neg p$ are in the set. But we know from (A7) that $Cp \rightarrow p$. Thus, if we have Cp we have p , but we also have $\neg p$: contradiction. Thus, Γ_1 is not consistent as it can prove a negation of something that is in the set (i.e., it does *not* hold that $\mathbf{KEC}_{(m)} \not\vdash \neg(\wedge_{\Psi \in \Gamma} \Psi)$).

Γ_2 is a maximally $\mathbf{KEC}_{(m)}$ -consistent set in Φ . This is because it is $\mathbf{KEC}_{(m)}$ -consistent, i.e., $\mathbf{KEC}_{(m)} \not\vdash \neg(\wedge_{\Psi \in \Gamma} \Psi)$ and because there is no $\Gamma' \subseteq \Phi$ such that $\Gamma_2 \subset \Gamma'$ and Γ' is still $\mathbf{KEC}_{(m)}$ -consistent.

Γ_3 is not a maximally $\mathbf{KEC}_{(m)}$ -consistent set in Φ . From Axiom (A8) we have $Cp \rightarrow ECp$. Since we have Cp in the set we can derive ECp . But we also have $\neg ECp$ in the set. Thus, Γ_3 is not consistent as it can prove a negation of something that is in the set (i.e., it does *not* hold that $\mathbf{KEC}_{(m)} \not\vdash \neg(\wedge_{\Psi \in \Gamma} \Psi)$).

b